

Perturbations divisible by u^k do not change the leading term:

$$Z(a + u^k \tilde{J}) \sim Z(a) \sim C n^{-p/2} (\log n)^m$$

Claude, with Daniel Murfet

2026-09-06

Abstract

The first multiplicity result for non-constant ξ . If $\xi(u) = a + u^k \tilde{J}(u)$ with $\tilde{J}$ continuous on the box and $|\tilde{J}| \leq L$, then with $t = \sqrt{n} u^k$ the exponent is $\beta t a + \beta t^2 \tilde{J} / \sqrt{n}$ and the one-term exponential remainder bound gives, for $n \geq 16L^2$, $|Z(\xi) - Z(a)| \leq \frac{4L}{\sqrt{n}} Z_{\beta/2, 2a}(n)$. The right-hand side has the same order $n^{-p/2} (\log n)^m$ as $Z(a)$ by unit 59 at $(\beta/2, 2a)$, so $Z(\xi) \sim Z(a)$ and the multiplicity theorem holds verbatim for such ξ . Zero `sorry/axiom`.

1 Setting

The Taylor tree's higher layers are not individually lower order in general (a perturbation J that does not vanish on a minimal divisor component contributes at leading order). Divisibility by u^k is the natural sufficient condition: every layer then carries an extra $n^{-p/2}$, and the whole tail is controlled at once by $|e^x - 1| \leq |x|e^{|x|}$ together with $x e^{-x} \leq 1$.

2 The things formalised

```
theorem exp_pert_abs_le ( a L t y : ) (h : 0 < ) (hL : 0 L) (h : 0 )
  (hy : |y| L) (hL : L * 1 / 4) :
  |Real.exp (- * t ^ 2 + * t * (a + t * y * )) - Real.exp (- * t ^ 2 + * t * a)|
  4 * L * * Real.exp (- ( / 2) * t ^ 2 + * t * a)
theorem boxIntegralXi_div_sub_le ... (hL : 16 * L ^ 2 n) :
  |boxIntegralXi a b (Real.sqrt n) k h (fun u => ( i, u i ^ k i) * J u)
  - boxIntegralFin a b (Real.sqrt n) k h|
  4 * L / Real.sqrt n * boxIntegralFin ( / 2) (2 * a) b (Real.sqrt n) k h
theorem boxIntegralXi_div_isEquivalent ...
  (J : (Fin (D + 2) → ) → ) (hJ : Continuous J) :
  (p : ) (m : ) (C : ), 0 < C ( i, p finExp k h i)
  m + 1 = (Finset.univ.filter fun i => finExp k h i = p).card
  (fun n => boxIntegralXi a b (Real.sqrt n) k h (fun u => ( i, u i ^ k i) * J u))
  ~[atTop] fun n => C * (n ^ -(p / 2)) * Real.log n ^ m
```

3 Proof strategy

Pointwise, $e^{-\beta t^2 + \beta t a} |e^{\beta t^2 y \varepsilon} - 1| \leq e^{-\beta t^2 + \beta t a} \beta t^2 L \varepsilon e^{\beta t^2 / 4} = 4L \varepsilon e^{-(\beta/2)t^2 + \beta t a} \cdot \frac{\beta t^2}{4} e^{-\beta t^2 / 4}$ and the last factor is at most 1 (`Real.add_one_le_exp`). Integrating over the box (integrands continuous, hence

integrable; the pointwise bound holds almost everywhere by `ae_restrict_mem`) gives the perturbation bound. For the asymptotic, unit 59 is applied at (β, a) and at $(\beta/2, 2a)$; the two minima and multiplicities coincide (both are read off from (k, h)), the difference is $O(n^{-1/2}Z_{\beta/2, 2a}) = o(Z(a))$ via `IsLittle0.mul_isBig0` with $n^{-1/2} = o(1)$, and `IsEquivalent.add_isLittle0` concludes.

4 Roadblocks and resolutions

- A `set` of $t = \sqrt{n} \prod u^k$ before unfolding the integrands did not fold the later-introduced occurrences, and a subsequent rewrite of $\prod u^k$ hit the inside of t ; introduce t as an opaque variable after unfolding and rewrite $\sqrt{n} \prod u^k$ into it.
- Nonnegativity of the box integral needs the almost-everywhere box membership (`integral_nonneg_of_ae`), since $\prod u_i^{h_i}$ is not globally nonnegative.
- `isLittle0_one_iff` takes the codomain type explicitly.

5 What was Mathlib, what was new

Mathlib: `Real.add_one_le_exp`, `Real.exp_le_exp`, `norm_integral_le_of_norm_le`, `integral_nonneg_of_ae`, `isLittle0_one_iff`, `tendsto_rpow_neg_atTop`, `IsLittle0.mul_isBig0`, `IsEquivalent.add_isLittle0`. New: the perturbation bound and the multiplicity theorem for $\xi = a + u^k \tilde{J}$.

6 Lessons

Comparing $Z(\xi)$ with a constant- ξ integral at a *different* temperature $(\beta/2, 2a)$ is the trick that makes the remainder uniform: the extra $e^{\beta t^2/4}$ from the perturbation is absorbed into the kernel, and the asymptotic theory for constant ξ is available at every temperature.

7 Outcome

The multiplicity theorem now covers a genuine class of non-constant ξ : those whose variation is divisible by the resolution monomial u^k . Perturbations that do not vanish on the minimal divisor components remain part of the leading coefficient (the state-density picture of the paper).